



CSI WESLEY

INSTITUTE OF TECHNOLOGY AND SCIENCES

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UNDER THE GUIDANCE OF

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DEPT. OF CSE

BATCH NO : 2021 – 2025

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Early Detection of Cardiac Arrest in Newborns Using Statistical Machine Learning in Cardiac ICU

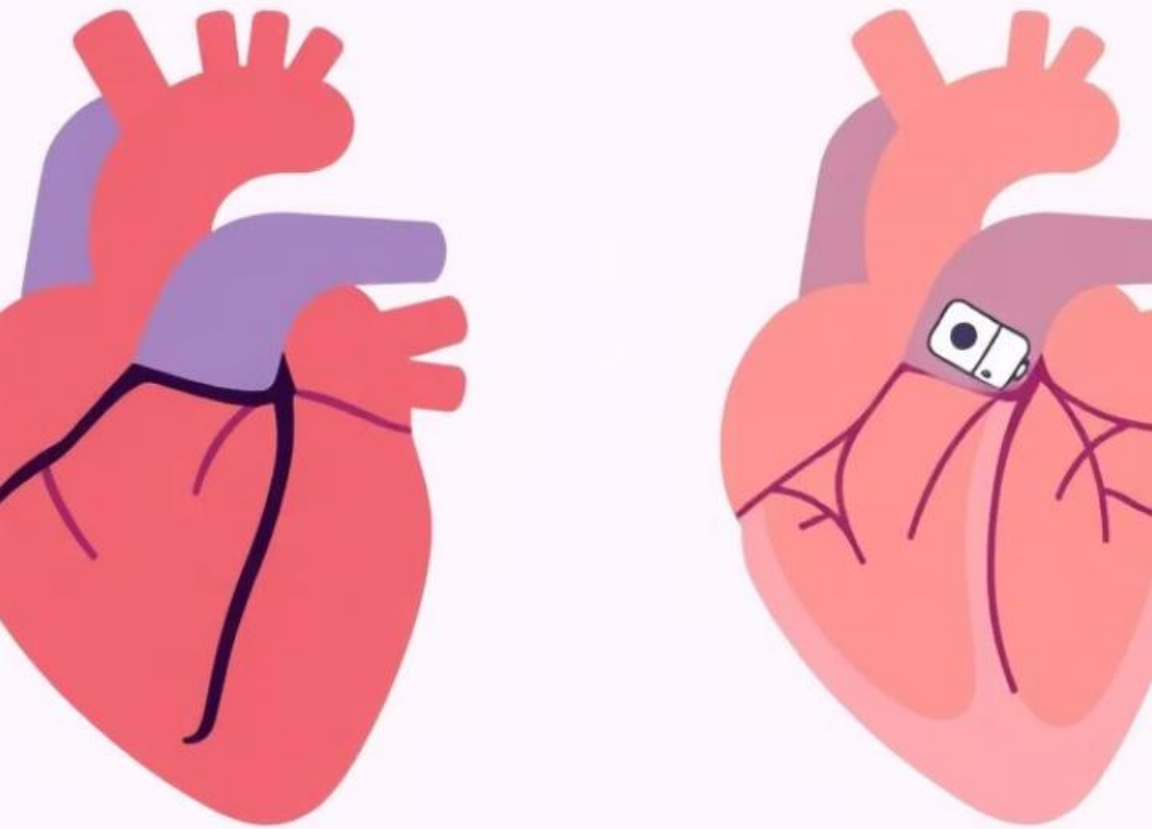


Introduction

Cardiac arrest in Newborns is a life-threatening condition that can cause serious complications or death. Early detection is crucial for ensuring timely treatment and better outcomes. Key signs include a rapid heart rate, difficulty breathing, bluish skin (cyanosis), unresponsiveness, and reduced movement. These symptoms require immediate medical attention.

Several risk factors can increase the likelihood of cardiac arrest in Newborns, such as low birth weight, preterm birth, difficult delivery, a family history of cardiac issues, and maternal high blood pressure during pregnancy. Reviewing the baby's medical history is also important to assess potential risks.

To detect cardiac arrest early, regular monitoring of heart rate and respiratory rate is essential. Pulse oximetry, a non-invasive and painless method, is commonly used to measure the oxygen level in the baby's blood, helping identify early signs of distress and preventing serious outcomes.



Existing Systems

Carlisle et al. [21] discusses heart failure, a condition where the heart can't pump enough blood due to causes like high blood pressure, coronary artery disease, and diabetes. Atrial fibrillation, an irregular heartbeat, is a common cause, leading to symptoms such as shortness of breath and fatigue. Treatment typically includes medications, lifestyle changes, and sometimes surgery.



Proposed System

The proposed system uses machine learning to detect cardiac arrest in newborns early by analyzing vital signs like heart rate, breathing, and oxygen levels. ML models can identify subtle changes that may not be visible to the naked eye. These models predict risk factors and provide alerts hours before traditional methods. This allows timely medical intervention, increasing survival chances and reducing complications. The system also helps identify high-risk infants for closer monitoring. ML enables continuous, non-invasive monitoring and personalized care based on patient data. Overall, it improves detection accuracy and supports better neonatal health outcomes.



Methodology

Data Collection & preparation

Source: Clinical ICU dataset for newborns

Features: Age, Chest pain type, Blood Pressure, Heart Rate, ECG, Oxygen levels.

Preprocessing: Handle missing values, Convert text to numbers.

1

Model Development

Algorithms : Logistic Regression,

Random Forest, SVM(support vector machine).

Training & Testing: Split data into 80% training & 20% testing

Evaluation : Accuracy, Precision,

Recall.

2

Prediction Flow

Input : Newborn babies health parameters.

Processing:

Preprocess —> predict using model.

Output: Predict risk of Cardiac arrest: Yes/No.

Help doctors with early warning.

3

System Architecture

Frontend (User Input)—> : Django —>ML model—> output.

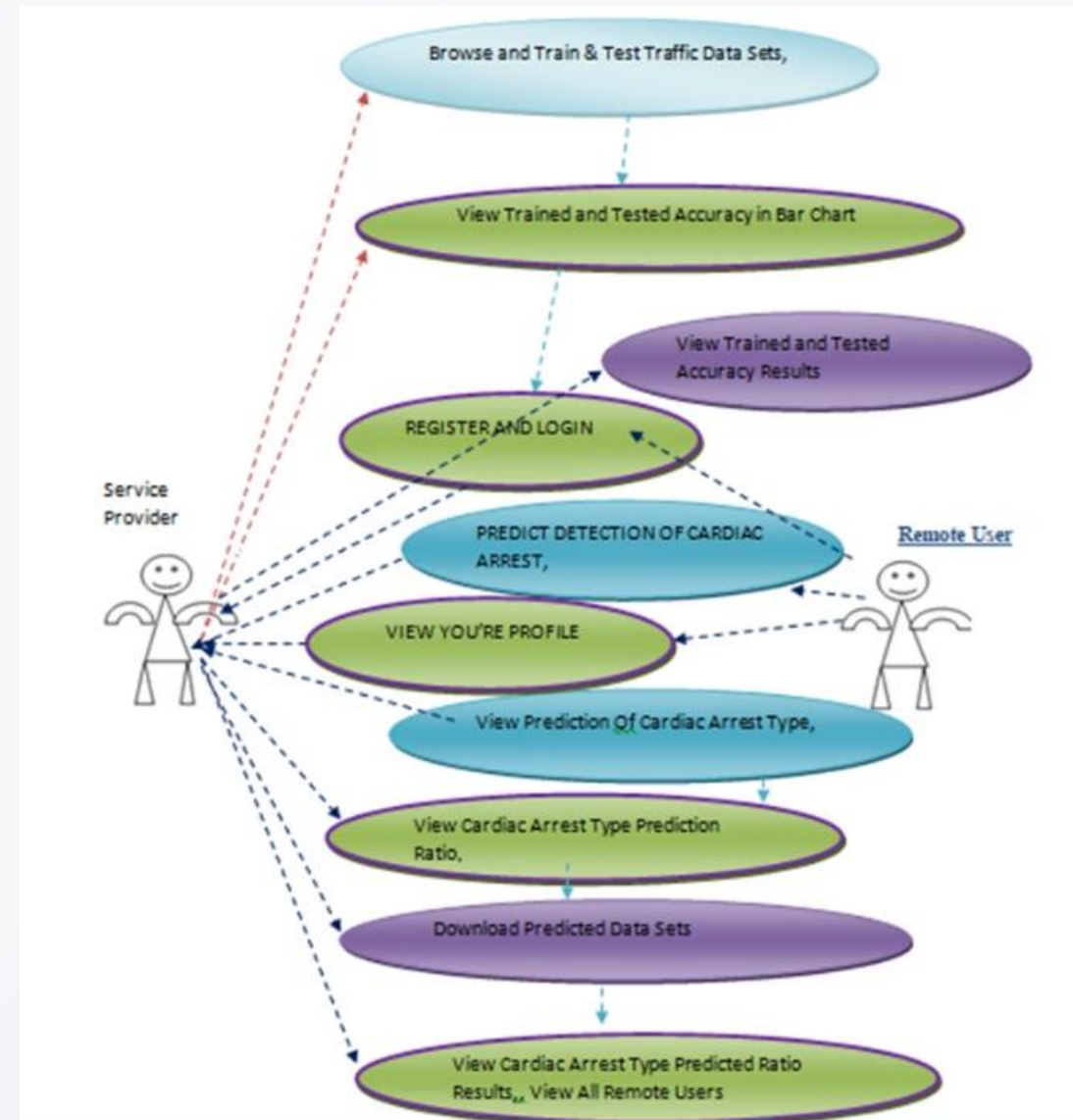
4

Deployment & Maintenance

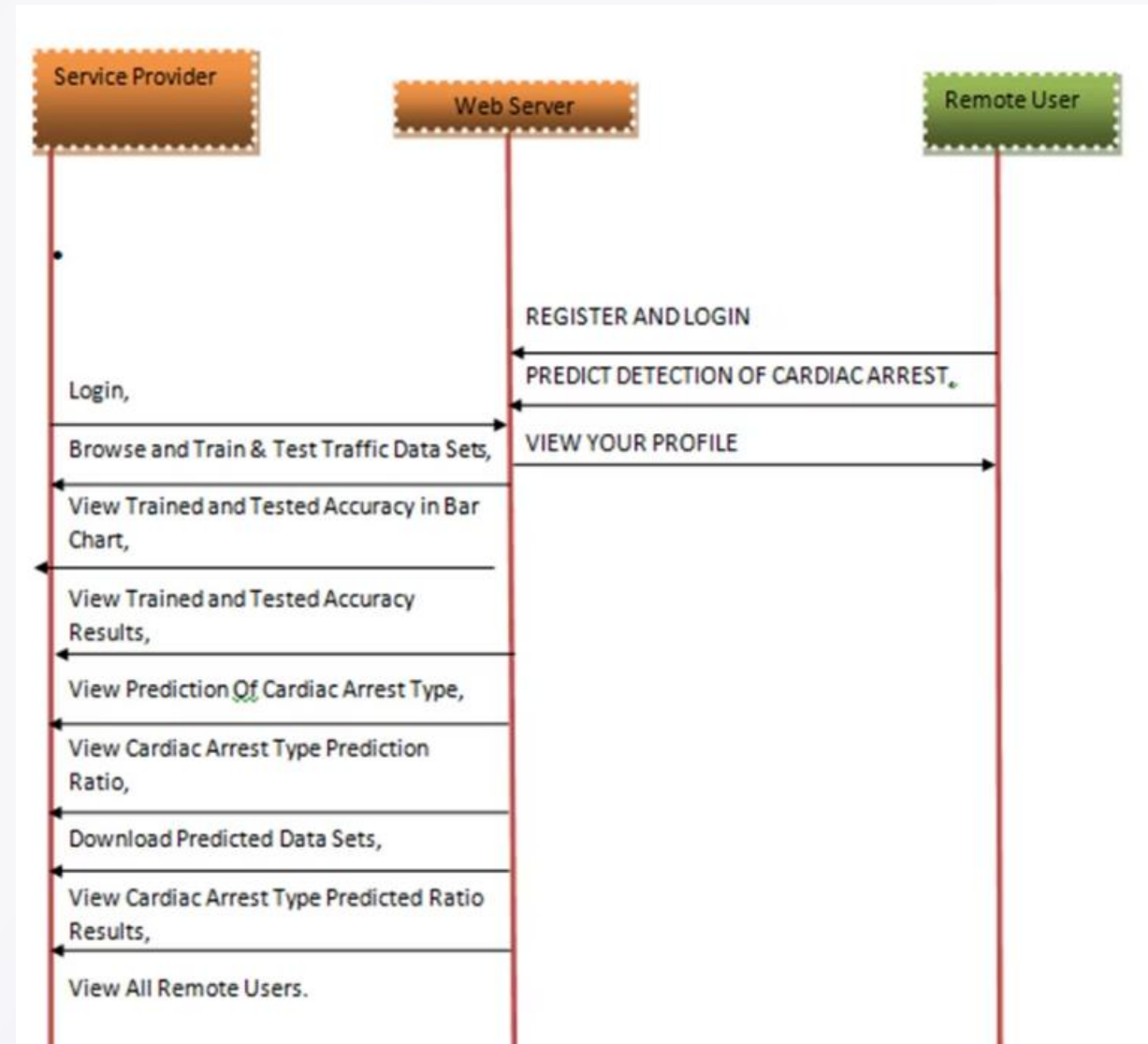
Flask : Web app (local / server).

5

Use Case Diagram



Sequence Diagram





Software Requirements

Software Requirements

- Operating system : Windows 7 Ultimate.
- Coding Language : Python.
- Front-End : Python.
- Back-End : Django-ORM
- Designing : Html, CSS, JavaScript.
- Data Base : MySQL (WAMP Server).

Hardware Requirements

1. Processor - Pentium –IV
2. RAM - 4 GB (min)
3. Hard Disk - 20 GB
4. Network - stable internet connection



Test Cases

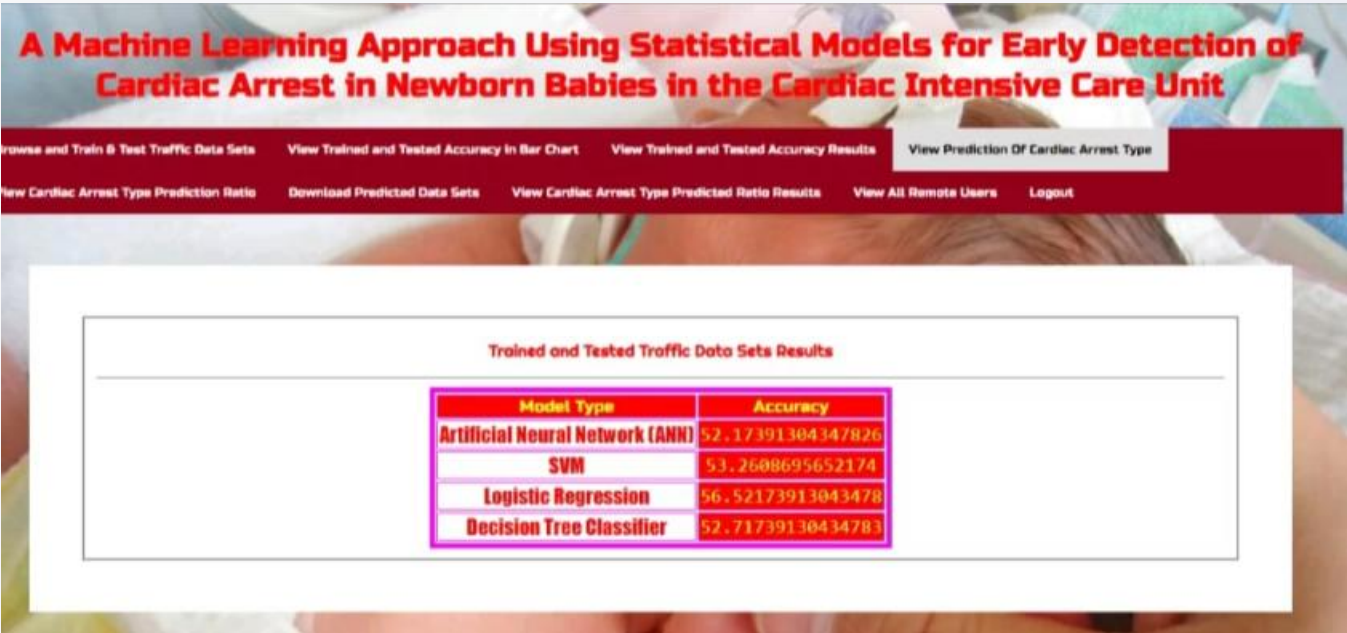
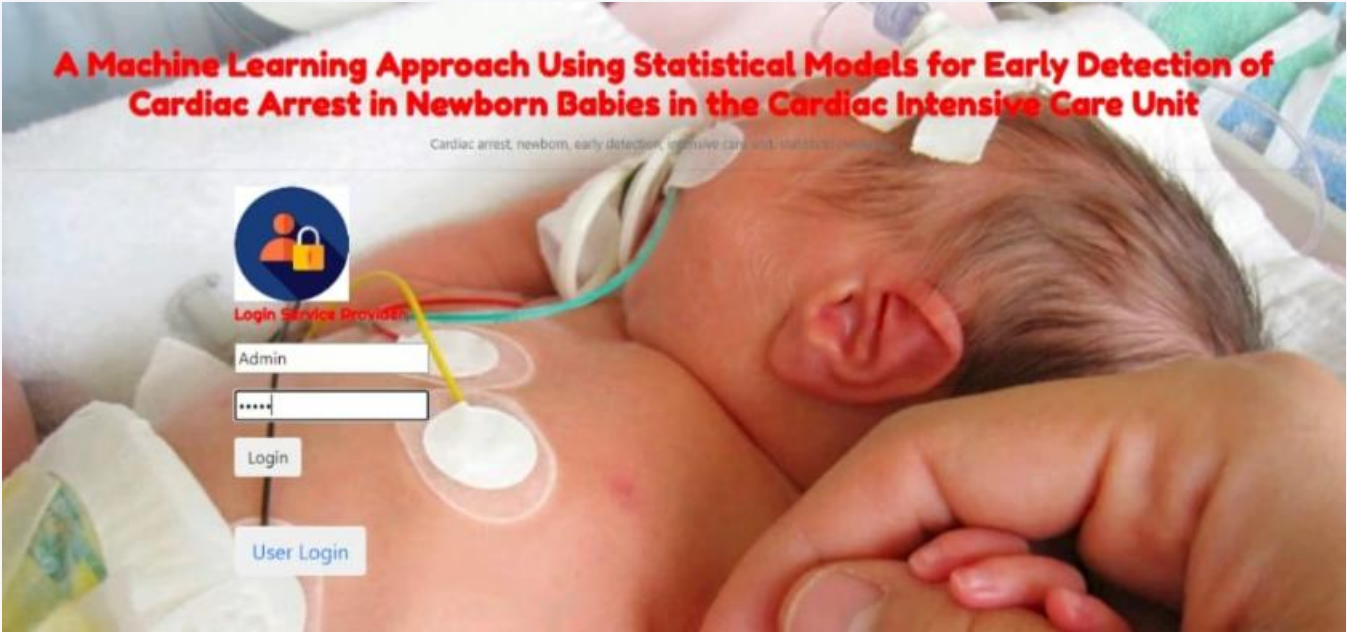
Test Case ID	Test Scenario	Input	Output
TC_01	Login with valid admin credentials	Username : Admin, password : ***	Redirect to admin dashboard
TC_02	Login with invalid password	Username: Admin, password wrong	Show error: Invalid Credentials
TC_03	Register with valid user details	All required fields	Registration success and login allowed
TC_04	Login with empty fields	Blank username/ password	show validation error
TC_05	Login with registered User	Email : www@gmail.com, password: 1234	Redirect to user dashboard

Test Case ID	Test Scenario	Input	Output
TC-06	Normal vital signs	HR=130, SpO2=98%	No alert
TC_07	Low Oxygen Saturation	SpO2=84%	Alert : Risk of cardiac arrest
TC_08	Enter extreme values	Age: 999,Heart Rate: 500	Handle gracefully or show warning
TC_09	All vital signs in dangerous range	HR=180,Spo2=75%	High Risk Alert

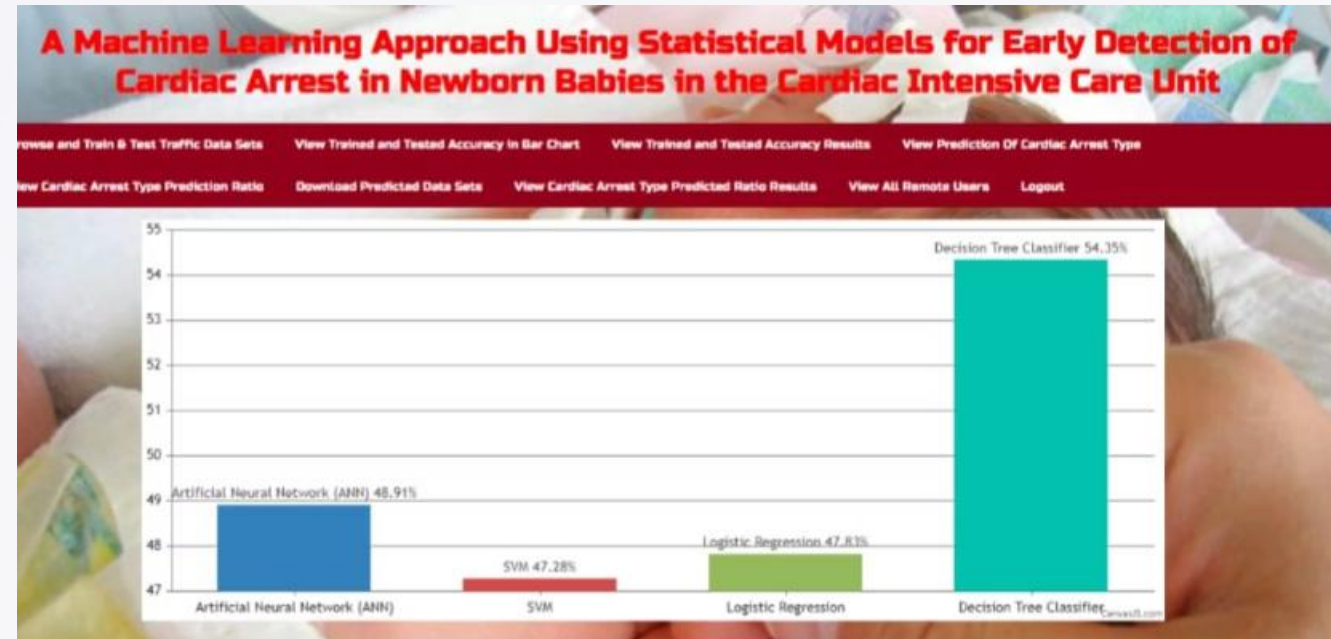
Interface



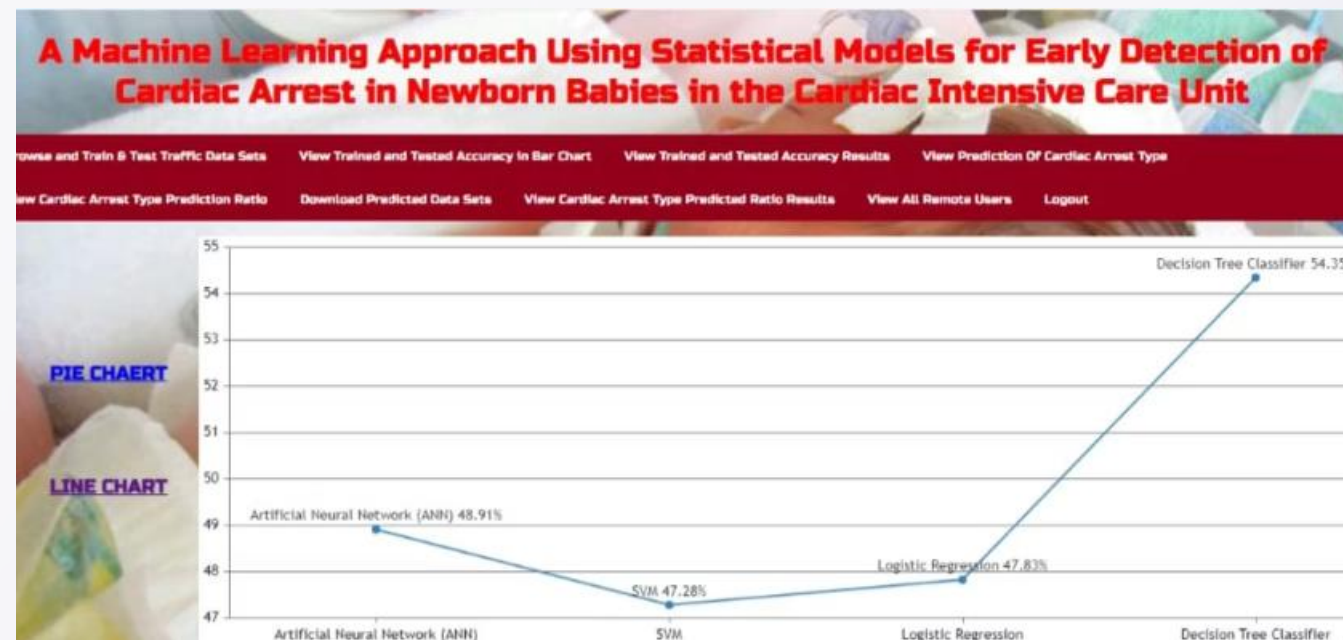
service providers



Trained and Test Accuracy In Bar Chart



View Trained and Test Accuracy Results



View Prediction Of Cardiac Arrest Type

rowse and Train & Test Traffic Data SetsView Trained and Tested Accuracy in Bar ChartView Trained and Tested Accuracy ResultsView Prediction Of Cardiac Arrest Type

View Cardiac Arrest Type Prediction RatioDownload Predicted Data SetsView Cardiac Arrest Type Predicted Ratio ResultsView All Remote UsersLogout

View Detection of Cardiac Arrest in Newborn Babies Details !!!

Fid	Age_In_Days	Sex	ChestPainType	RestingBP-Heart Rate	RestingECG	MaxHR	ExerciseAngine	Oldpeak	ST_Slope	slp	caa	thall	Pre
172.217.10.142-10.42.0.151-443-51582-6	54	M	ATA	110	Normal	142	N	0	Up	2	1	3	No Cardiac Arrest Found
10.42.0.211-10.42.0.1-1420-53-17	49	M	ASY	140	Normal	140	Y	1	Flat	1	0	2	Cardiac Arrest Found
172.217.10.227-10.42.0.151-443-34159-6	42	F	NAP	115	ST	137	N	0	Up	2	0	2	No Cardiac Arrest Found

View cardiac Arrest Type Prediction Ratio

A Machine Learning Approach Using Statistical Models for Early Detection of Cardiac Arrest in Newborn Babies in the Cardiac Intensive Care Unit

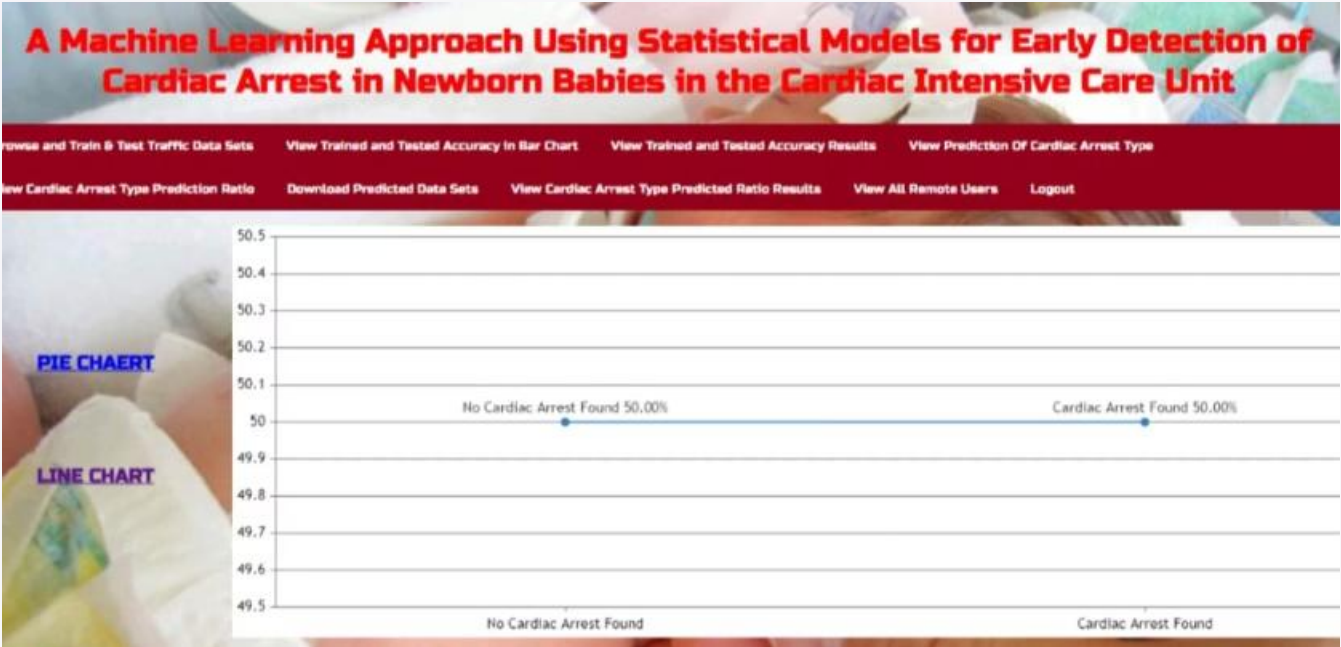
rowse and Train & Test Traffic Data SetsView Trained and Tested Accuracy in Bar ChartView Trained and Tested Accuracy ResultsView Prediction Of Cardiac Arrest Type

View Cardiac Arrest Type Prediction RatioDownload Predicted Data SetsView Cardiac Arrest Type Predicted Ratio ResultsView All Remote UsersLogout

View Detection of Cardiac Arrest in Newborn Babies Type Found Ratio Details

Cardiac Arrest Type	Ratio
No Cardiac Arrest Found	50.0
Cardiac Arrest Found	50.0

View Cardiac Arrest Type Predicted Ratio Results



View All Remote Users

A Machine Learning Approach Using Statistical Models for Early Detection of Cardiac Arrest in Newborn Babies in the Cardiac Intensive Care Unit

View Trained and Tested Accuracy In Bar Chart View Trained and Tested Accuracy Results View Prediction Of Cardiac Arrest Type

View Cardiac Arrest Type Prediction Ratio Download Predicted Data Sets View Cardiac Arrest Type Predicted Ratio Results **View All Remote Users** Logout

VIEW ALL REMOTE USERS !!!

USER NAME	EMAIL	Mob No	Country	State	City
Gopalan	Gopalan123@gmail.com	9535866270	India	karnataka	Bangalore
Manjunath	tmksmanju14@gmail.com	9535866270	India	Karnataka	Bangalore
www	www@gmail.com	9988776655	Ind	Telangana	hyd



User Registration



User Login

A Machine Learning Approach Using Statistical Models for Early Detection of Cardiac Arrest in Newborn Babies in the Cardiac Intensive Care Unit

Cardiac arrest, newborn, early detection, intensive care unit, statistical models

 Login Using Your Account

Login Using Your Account

A Machine Learning Approach Using Statistical Models for Early Detection of Cardiac Arrest in Newborn Babies in the Cardiac Intensive Care Unit

PREDICT DETECTION OF CARDIAC ARREST VIEW YOUR PROFILE LOGOUT

YOUR PROFILE DETAILS !!!

USER NAME = www

EMAIL = www@gmail.com

PASSWORD = 1234

MOBILE NO = 9988776655

COUNTRY = Ind

STATE = Telangana

CITY = hyd

Prediction Of Cardiac Arrest Type

PREDICTION OF CARDIAC ARREST TYPE III

ENTER ALL DETAILS FROM HEALTHCARE DATA SETS III

ENTER FId

192.229.163.180-10.42.0.42

ENTER Sex

M

ENTER RestingBP(Heart Rate)

140

ENTER MaxHR

172

ENTER Oldpeak

0

ENTER slp

1

ENTER thall

1

ENTER Age_In_Days

40

ENTER ChestPainType

ATA

ENTER RestingECG

Normal

ENTER ExerciseAngina

N

ENTER ST_Slope

Up

ENTER cse

0

Predict

Prediction of Cardiac Arrest Type

=

ENTER ALL DETAILS FROM HEALTHCARE DATA SETS III

ENTER FId

ENTER Sex

--Select--

ENTER RestingBP(Heart Rate)

ENTER MaxHR

ENTER Oldpeak

ENTER slp

ENTER thall

ENTER Age_In_Days

ENTER ChestPainType

ENTER RestingECG

ENTER ExerciseAngina

ENTER ST_Slope

ENTER cse

Predict

Prediction of Cardiac Arrest Type

=

No Cardiac Arrest Found

Conclusion

The proposed machine learning model accurately detects early signs of cardiac arrest in newborns using vital sign data. It enables timely intervention, improving survival rates and reducing CICU stays. Future enhancements include real-time data use and AI-driven personalized care.



Thank You